WORKED SOLUTION

STEP II 1991, Question 4

Pure • Trigonometry

STEP II 1991, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/91-s2-q4

This question has a full guided treatment in *Beyond STEP*, Section 6.2, *Gauss's 17-gon*: https://beyondstep.keithmcnulty.org/ch06/gauss-17gon.html

The key idea

Because $5\phi = 2\pi$ and $17\theta = 2\pi$, cosines of large multiples of the angle fold back onto small ones: $\cos(n-k)\phi = \cos k\phi$ when $n\phi = 2\pi$. So when we square or multiply sums of cosines with the product formulae, everything collapses back into the same few cosines. In the second half we split the eight cosines $\cos\theta, \ldots, \cos 8\theta$ into two groups of four, find their sum and their product, and so get a quadratic of which z is a root.

Solution

The pentagon: $\phi = 2\pi/5$

We use the product formula $2\cos A\cos B = \cos(A+B) + \cos(A-B)$, together with $\cos^2 A = \frac{1}{2}(1+\cos 2A)$. Since $5\phi = 2\pi$,

$$\cos 3\phi = \cos(2\pi - 2\phi) = \cos 2\phi, \qquad \cos 4\phi = \cos(2\pi - \phi) = \cos\phi.$$

Hence

$$\begin{aligned} y^2 &= \cos^2\phi + 2\cos\phi\cos 2\phi + \cos^2 2\phi \\ &= \tfrac{1}{2}(1+\cos 2\phi) + (\cos 3\phi + \cos\phi) + \tfrac{1}{2}(1+\cos 4\phi) \\ &= \tfrac{1}{2}(1+\cos 2\phi) + (\cos 2\phi + \cos\phi) + \tfrac{1}{2}(1+\cos\phi) \\ &= 1 + \tfrac{3}{2}(\cos\phi + \cos 2\phi) = 1 + \tfrac{3}{2}y. \end{aligned}$$

Multiplying by 2 gives $2y^2 = 3y + 2$, as required.

This factorises as $(2y+1)(y-2) = 0$, so $y = -\frac{1}{2}$ or $y = 2$. But $2\phi = \frac{4\pi}{5}$ lies between $\frac{\pi}{2}$

and π, so $\cos 2\phi < 0$, while $\cos\phi \le 1$. Thus $y < 1$, which rules out $y = 2$, and so

$$y = \cos\tfrac{2\pi}{5} + \cos\tfrac{4\pi}{5} = -\tfrac{1}{2}.$$

The sum of the cosines for $\theta = 2\pi/17$

Let $\omega = \cos\theta + \mathrm{i}\sin\theta$. By de Moivre's theorem $\omega^k = \cos k\theta + \mathrm{i}\sin k\theta$, and $\omega^{17} = \cos 2\pi + \mathrm{i}\sin 2\pi = 1$, while $\omega \neq 1$. Summing the geometric series,

$$\sum_{k=0}^{16} \omega^k = \frac{\omega^{17} - 1}{\omega - 1} = 0.$$

Taking real parts gives $\displaystyle\sum_{k=0}^{16} \cos k\theta = 0$, as required.

The value of z

Write $c_k = \cos k\theta$. Since $17\theta = 2\pi$ and cosine is even,

$$c_{17-k} = c_k \quad \text{and} \quad c_{-k} = c_k,$$

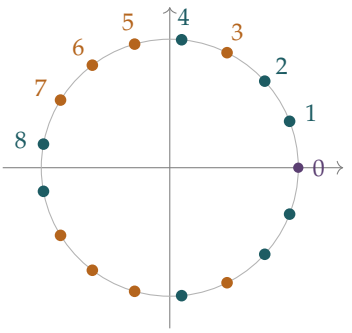
so every c_k equals one of $c_1, c_2, \ldots, c_8$. In particular the sum just proved, with $c_0 = 1$ removed, reads $2(c_1 + c_2 + \cdots + c_8) = -1$, so

$$c_1 + c_2 + \cdots + c_8 = -\tfrac{1}{2}. \qquad (*)$$

The angles in z are $1, 2, 4, 8$ (each double the last, reduced as above: $16 \to 1$). The other four, $3, 5, 6, 7$, give the companion sum

$$w = c_3 + c_5 + c_6 + c_7.$$

The figure shows the seventeen points ω^k on the unit circle: the teal ones (and their mirror images) carry the cosines in z, the amber ones those in w.



The sum. By $(*)$, $z + w = c_1 + c_2 + \cdots + c_8 = -\tfrac{1}{2}$.

The product. Expanding zw gives sixteen products $c_a c_b$ with $a \in \{1,2,4,8\}$ and $b \in \{3,5,6,7\}$, and each is $\frac{1}{2}(c_{a+b} + c_{a-b})$. Reducing each index to the range 1 to 8 (using $c_{-k} = c_k$ and $c_{17-k} = c_k$), the pairs of indices are:

	$b=3$	$b=5$	$b=6$	$b=7$
$a=1$	4, 2	6, 4	7, 5	8, 6
$a=2$	5, 1	7, 3	8, 4	8, 5
$a=4$	7, 1	8, 1	7, 2	6, 3
$a=8$	6, 5	4, 3	3, 2	2, 1

(For example $a=8$, $b=7$ gives c_{15} and c_1, and $c_{15} = c_2$.)

? **Ask yourself** Do we need to know which cosines appear, or only how often?

Only how often. Counting the table, each of the indices $1, 2, \ldots, 8$ appears exactly four times among the 32 entries. So

$$zw = \tfrac{1}{2} \cdot 4\,(c_1 + c_2 + \cdots + c_8) = 2 \cdot \left(-\tfrac{1}{2}\right) = -1,$$

using $(*)$ again.

The quadratic. Since $z + w = -\frac{1}{2}$ and $zw = -1$, the numbers z and w are the roots of

$$t^2 + \tfrac{1}{2}t - 1 = 0, \qquad \text{that is,} \qquad 2t^2 + t - 2 = 0,$$

so $\{z, w\} = \left\{ \dfrac{-1+\sqrt{17}}{4},\ \dfrac{-1-\sqrt{17}}{4} \right\}$. The first root is positive and the second negative.

Choosing the root. We show $z > 0$. Since $\frac{4}{17} < \frac{1}{4}$, both θ and 2θ lie in $(0, \frac{\pi}{4})$, so c_1 and c_2 both exceed $\frac{1}{\sqrt{2}}$ and $c_1 + c_2 > \sqrt{2} > 1$. Also $4\theta = \frac{8\pi}{17} < \frac{\pi}{2}$, so $c_4 > 0$, and $c_8 \geq -1$. Therefore

$$z = (c_1 + c_2) + c_4 + c_8 > 1 + 0 - 1 = 0,$$

and z must be the positive root:

$$z = \frac{-1+\sqrt{17}}{4} = -\frac{1-\sqrt{17}}{4}.$$

Checks

Numerically, $\cos 72^\circ + \cos 144^\circ = 0.30902 - 0.80902 = -0.5$. With $\theta = 2\pi/17$, a calculator gives $z = 0.932472 + 0.739009 + 0.092268 - 0.982973 = 0.780776$, and $(\sqrt{17} - 1)/4 = 0.780776$; also $w = -1.280776 = (-1 - \sqrt{17})/4$, so $z + w = -0.5$ and $zw = -1$, as found.

Common slips

- Forgetting to rule out $y = 2$ in the first part, or w's root in the second: a quadratic gives two candidates, and a separate sign argument is needed to pick one.
- Including $k = 0$ in $(*)$: the full sum is 0, but $c_0 = 1$, so $c_1 + \cdots + c_{16} = -1$, and halving gives $-\frac{1}{2}$.
- Mis-reducing indices in the product table, such as treating c_9 as new rather than as c_8.

Beyond STEP

This is Gauss's famous calculation showing that the regular 17-gon can be constructed with ruler and compasses. Splitting the cosines by the powers of 3 (a generator modulo 17) into nested groups of 8, 4, 2 and 1 gives a chain of quadratics, so $\cos\frac{2\pi}{17}$ can be written using only square roots; z and w are the first stage of that chain, called Gaussian periods. The full story belongs to Galois theory: see Allenby, *Rings, Fields and Groups*, and, for the arithmetic of roots of unity, Davenport, *The Higher Arithmetic*.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.